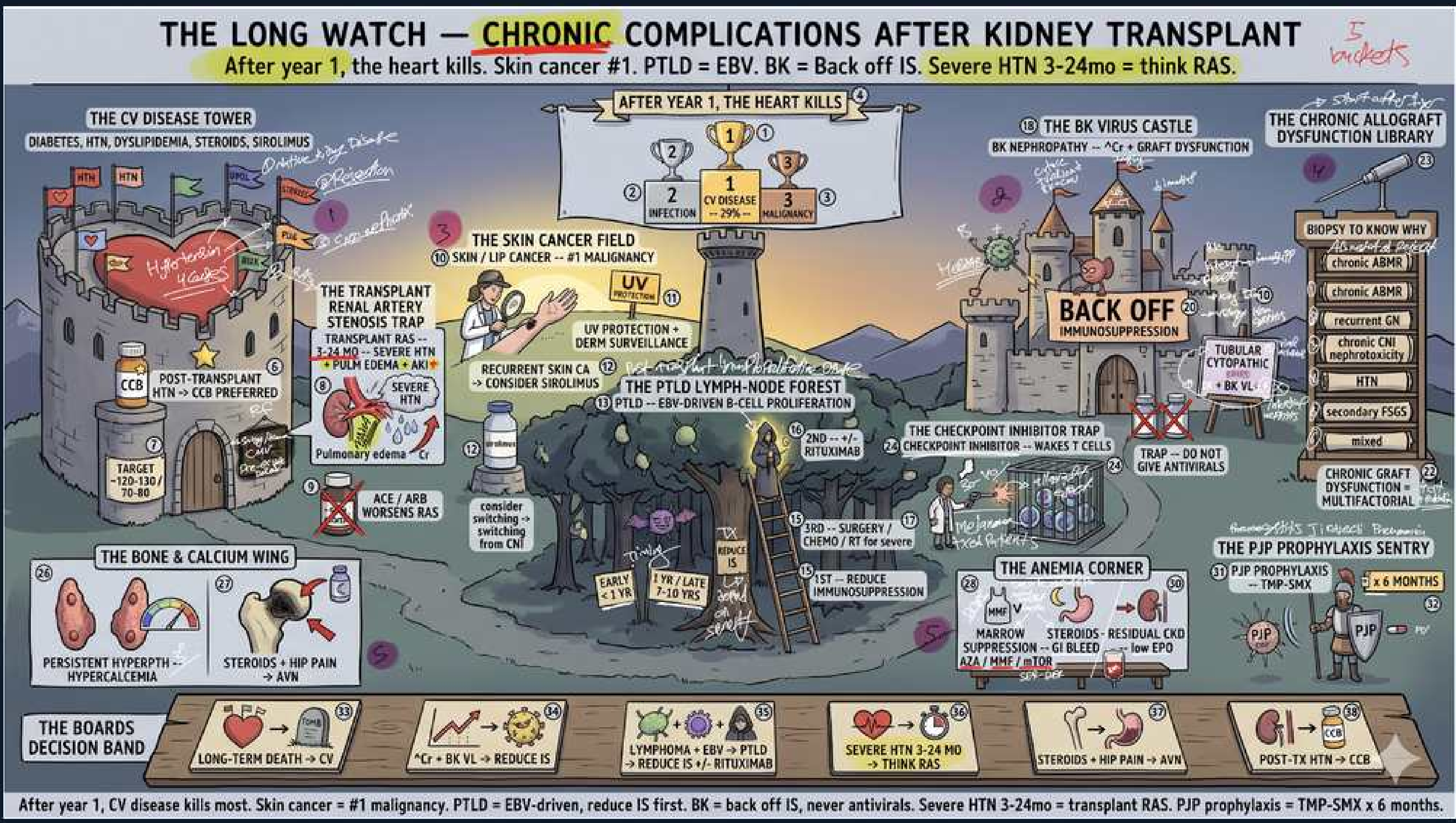
Chronic Complications After Kidney Transplant (After Year 1)



1. After year 1, heart kills

WHAT YOU SEE
Banner: heart on the castle keep

WHAT IT ENCODES
Cardiovascular disease is the #1 cause of death in kidney transplant recipients with a functioning graft, especially after the first year. Death-with-functioning-graft is dominated by CV events, not rejection. Aggressive control of HTN, lipids, and diabetes is mandatory.

BOARD TRAP
Calling infection or acute rejection the leading late cause of death — those dominate the FIRST year; after year 1 it is cardiovascular disease.

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2. CV disease risk drivers

WHAT YOU SEE
CV disease tower listing DM, HTN, dyslipidemia, steroids, sirolimus

WHAT IT ENCODES
Immunosuppression itself accelerates CV risk: steroids and CNIs cause HTN/dyslipidemia/new-onset diabetes (NODAT), and sirolimus (mTOR inhibitor) causes hypertriglyceridemia. Manage as a very-high-risk CV patient with statins and tight BP control.

BOARD TRAP
Attributing post-transplant dyslipidemia only to diet — sirolimus and steroids are major drug causes and may prompt regimen change.

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3. PTLD = EBV-driven

WHAT YOU SEE
Lymph-node forest, EBV label, 'PTLD'

WHAT IT ENCODES
Post-transplant lymphoproliferative disorder is driven by EBV in the setting of T-cell suppression, usually B-cell lineage. First-line treatment is REDUCTION of immunosuppression; add rituximab (anti-CD20) for CD20+ disease, chemo if needed. Highest risk in EBV D+/R- recipients.

BOARD TRAP
Reaching for aggressive chemo first — initial step in PTLD is reducing immunosuppression, then rituximab for CD20+ B-cell disease.

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4. Skin cancer #1 malignancy

WHAT YOU SEE
Skin cancer field, SCC/lip cancer

WHAT IT ENCODES
Skin cancer is the most common malignancy after solid-organ transplant; squamous cell carcinoma predominates (reversed SCC:BCC ratio vs general population) and is more aggressive. Counsel sun protection and dermatologic surveillance; consider switching to sirolimus, which lowers skin-cancer risk.

BOARD TRAP
Assuming basal cell carcinoma is most common — in transplant patients SCC outnumbers BCC and behaves more aggressively.

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5. BK virus nephropathy

WHAT YOU SEE
BK virus castle, tubular cytopathic cell

WHAT IT ENCODES
BK polyomavirus nephropathy causes graft dysfunction and rising creatinine, diagnosed by plasma BK PCR and biopsy showing tubular viral cytopathic changes / decoy cells. Treatment is REDUCING immunosuppression — there is no reliable antiviral. BK behaves opposite to rejection.

BOARD TRAP
Treating rising Cr with MORE immunosuppression when biopsy shows BK — that worsens it; BK is managed by REDUCING immunosuppression.

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6. Transplant renal artery stenosis

WHAT YOU SEE
Renal artery stenosis trap, severe HTN 3-24mo

WHAT IT ENCODES
Transplant renal artery stenosis presents as severe or refractory hypertension 3–24 months post-transplant, sometimes with graft dysfunction and ACE/ARB-induced creatinine rise. Diagnose with Doppler/angiography; treat with angioplasty +/- stent.

BOARD TRAP
Assuming new severe HTN months after transplant is essential HTN — think transplant RAS, especially if ACE/ARB worsens the graft Cr.

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7. ACE/ARB worsens RAS

WHAT YOU SEE
ACE/ARB pill with worsening arrow

WHAT IT ENCODES
In transplant renal artery stenosis, the graft is a single functional kidney; ACE inhibitors/ARBs drop efferent tone and can precipitate an acute creatinine rise. A sharp Cr jump after starting ACE/ARB is a clue to underlying RAS.

BOARD TRAP
Continuing ACE/ARB despite a sharp Cr rise — in a solitary transplant kidney this signals RAS and the drug should be reconsidered.

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8. Checkpoint inhibitor caution

WHAT YOU SEE
Checkpoint inhibitor wakes T-cells, trap-do-not-give

WHAT IT ENCODES
Immune checkpoint inhibitors (anti-PD-1/CTLA-4) unleash T-cells and can trigger acute allograft rejection in transplant recipients. Use is high-risk and requires multidisciplinary discussion; rejection can be irreversible and cost the graft.

BOARD TRAP
Freely giving checkpoint inhibitors for a transplant patient's cancer — they reactivate T-cells and can precipitate graft rejection.

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9. Persistent hyperparathyroidism

WHAT YOU SEE

Bone & calcium wing, PTH

WHAT IT ENCODES

After transplant, tertiary (autonomous) hyperparathyroidism can persist, causing hypercalcemia and bone disease despite a functioning graft. Treat with cinacalcet or parathyroidectomy if persistent/symptomatic. Steroids add to bone loss.

BOARD TRAP

Expecting PTH to normalize immediately post-transplant — autonomous (tertiary) hyperparathyroidism with hypercalcemia can persist for months.

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10. Steroid-induced bone disease

WHAT YOU SEE

Steroids + hip pain icon

WHAT IT ENCODES

Chronic steroids cause osteoporosis and avascular (osteonecrosis) of the femoral head — new hip/groin pain in a transplant patient on steroids is AVN until proven otherwise (MRI). Steroid-sparing regimens and bone protection mitigate this.

BOARD TRAP

Dismissing hip pain in a steroid-treated transplant patient as arthritis — MRI for avascular necrosis (osteonecrosis) of the hip.

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11. Chronic allograft dysfunction

WHAT YOU SEE

Chronic allograft dysfunction library, biopsy

WHAT IT ENCODES

Chronic graft dysfunction (interstitial fibrosis/tubular atrophy) reflects cumulative immune and non-immune injury — chronic rejection, CNI toxicity, recurrent disease, BK, hypertension. Biopsy distinguishes causes and guides whether to adjust immunosuppression.

BOARD TRAP

Labeling all chronic graft loss as 'chronic rejection' — biopsy is needed because CNI toxicity, BK, and recurrent GN mimic it and are managed differently.

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12. CNI chronic nephrotoxicity

WHAT YOU SEE

Tubular cytopathic / CNI toxicity panel

WHAT IT ENCODES

Calcineurin inhibitors (tacrolimus, cyclosporine) cause dose-dependent afferent arteriolar vasoconstriction and chronic interstitial fibrosis, contributing to late graft decline. Trough levels and biopsy guide dose reduction or switch (e.g., to belatacept).

BOARD TRAP

Assuming a slow Cr creep on stable tacrolimus is always rejection — chronic CNI nephrotoxicity is a key differential and is managed by lowering the CNI.

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13. PJP prophylaxis sentry

WHAT YOU SEE

PJP prophylaxis sentry, TMP-SMX, 6 months

WHAT IT ENCODES

Pneumocystis jirovecii pneumonia prophylaxis with TMP-SMX is given for at least 6 months post-transplant (often longer). TMP-SMX also covers Toxoplasma, Nocardia, and UTIs; dapsone/atovaquone for sulfa allergy.

BOARD TRAP

Stopping PJP prophylaxis too early — standard is at least 6 months of TMP-SMX, extended after rejection treatment or intensified immunosuppression.

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14. Anemia post-transplant

WHAT YOU SEE

Anemia corner, marrow + EPO

WHAT IT ENCODES

Post-transplant anemia stems from residual CKD (low EPO), marrow suppression from immunosuppressants (MMF, azathioprine, sirolimus, valganciclovir), and iron deficiency. Work up reversible causes before reflexively giving ESAs.

BOARD TRAP

Reflexively giving EPO for transplant anemia without checking drug-induced marrow suppression and iron studies first.

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15. PTLD lymph-node forest

WHAT YOU SEE

Lymphoma trees, B-cell, rituximab

WHAT IT ENCODES

PTLD often presents as lymphadenopathy or extranodal/GI masses; most are EBV+ B-cell lymphomas. Staging and biopsy confirm; CD20+ disease responds to rituximab after reducing immunosuppression. Early PTLD can regress with reduced immunosuppression alone.

BOARD TRAP

Missing extranodal/GI presentations of PTLD — it is not always nodal, and new GI symptoms in a transplant patient warrant evaluation.
